

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} - Xy - Yx$$

→ está completamente equilibrado

modelo constitutivo

$$\varepsilon_x = \frac{\sigma_x - \nu \sigma_y}{E}, \quad \varepsilon_y = \frac{\sigma_y - \nu \sigma_x}{E}, \quad \gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\nu)\tau_{xy}}{E}$$

condiciones de compatibilidad

$$\varepsilon_{xx} = \frac{1}{E} \left(\frac{\partial^2 \Phi}{\partial y^2} - \nu \frac{\partial^2 \Phi}{\partial x^2} \right)$$

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} = \frac{1}{E} \left(\frac{\partial^4 \Phi}{\partial y^4} - \nu \frac{\partial^4 \Phi}{\partial y^2 \partial x^2} \right)$$

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = 2 \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y}$$

$$\gamma_{xy} = 2 \varepsilon_{xy}$$

Ecuación biarmónica

Φ función de tensión de Airy

$$\nabla^4 \Phi = \frac{\partial^4 \Phi}{\partial y^4} + 2 \frac{\partial^4 \Phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \Phi}{\partial x^4} = 0$$

$$\Delta(\Delta \Phi) = \dots$$

c)

$$3a + 3e = -c$$

d)

$$\Phi(x, y) = ax^5 + bx^4y + cx^3y^2 + dx^2y^3 + exy^4 + fy^5 \rightarrow \nabla^4 \Phi = 0 \rightarrow$$

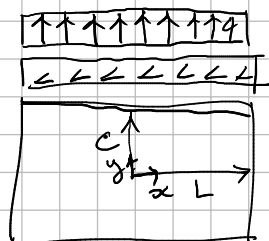
$$(5a + b + c).x + (5f + b + d).y = 0$$

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$$\Phi = \frac{a_2}{2} x^2 + b_2 xy + \frac{c_2}{2} y^2$$

$$\rightarrow \text{verifico que } \nabla^4 \Phi = 0 \rightarrow$$

como función prescrita → evalúa $\underline{T} = \underline{T} \cdot \underline{n}$ en la frontera.



$$T_x = c_2$$

$$T_y = a_2$$

$$T_{xy} = -b_2$$

$$5) \quad \nabla_x = c_3 x + d_3 y$$

$$\nabla_y = a_3 x + b_3 y$$

$$\nabla_{xy} = b_3 x + c_3 y$$